Computer Vision Peer Review

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1 Summary

This paper states a proposal for the development of a general purpose robot for assisting in the caretaking of elderly and disabled people. The goal of the paper is to develop a robot that can be adapted to tasks such as navigation, object recognition and avoidance, pose recognition, transport of hospital-related items, and communication. The paper then provides background and preliminary examples related to the area, citing two projects similar to the outcome proposed. It gives descriptions of these projects and discusses their solutions to related problems.

The paper then proposes the intended technical approach for building the robot to meet its criteria. The suggested implementation uses a Microsoft 360 Connect camera to detect and follow ArUco markers. It also opts for both a Raspberry Pi 3 and an Arduino Mega 2560 Microcontroller. For locomotion, the author decides to use four 4 degrees of freedom legs, with wheels attached to the ends of each of them. The author states that this will reduce power consumption while maintaining the ability for the robot to climb stairs. It then suggests how the robot may be evaluated and tested. It summarises potential test situations under which to evaluate the robot. These include pose recognition, motion detection, and outdoor navigation scenarios. It does not provide any experimental results or discussion of a completed outcome.

The author also attempts to address ethical concerns and potential future research directions. Ethical concerns are minimally touched on, with in-depth analysis stated to be "beyond the scope of" the paper. Some of the functionality discussed in the future work section overlaps with capabilities that are previously ambiguously described as already implemented. The section also generally details possible extensions for the robot's functionality, including point cloud maps, measuring the health of a user, and playing fetch. While the chosen application area is well motivated, the paper does not have sufficient technical depth, experimental results, or testing validation.

2 Decision

The final decision of the review is to reject this proposal. The paper requires major revisions to form it into a publishable state, due to the concerns outlined in the following sections. The paper has unclear contributions, insufficient novelty, and a lack of any completed experiments. Therefore, it does not meet the standards for publication of a computer vision research paper. Despite shortcomings in execution, the topic is well motivated and has the potential to form the basis of a compelling paper if developed further.

3 Major Concerns

A first major concern with this paper is the lack of clear contributions. It frequently mixes implemented functionality with speculative goals, making it difficult to understand what has been achieved. For example, capabilities such as outdoor navigation, pose detection, and object recognition are discussed despite not being experimentally demonstrated. This conflicts with the stated goal of developing a practical assistive robotic system, and it is unclear what the concrete research contributions of this paper are. Related to this concern, the evaluation has a lack of experimental or computational data. The entire evaluation and testing section is purely hypothetical and resembles speculative planning more than rigorous scientific evaluation, while also failing to demonstrate a novel technical contribution. In order to resolve this concern, the author would need to undergo extensive further work, including developing the proposed robot and conducting new experiments. This is a major factor in the decision to reject this paper. Revisions should include quantitative evaluation of a developed outcome, such as clearly defined success metrics, extensive testing, and benchmarks (e.g., for navigational performance or tracking accuracy).

Additionally, the design plan detailed in the paper lacks technical depth. Even if the outcome were created according to the specification, it would likely be inadequate in terms of novelty. It provides only a rough list of potential hardware that might be used to create the robot, but does not go into detail about the system design decisions made for the project. To resolve this concern, the approach section would need significant additions. For example, it should include discussion of the computer vision pipeline, navigational methodology, and system integration decisions.

Another concern is the minimal citation of previous work. Although there are a number of referenced papers in the bibliography, the author only addresses a couple of these. The references that are addressed are not evaluated in significant depth, and their relationship to the proposed system is often unclear.

Finally, the paper has an underdeveloped section on ethical and safety implications. It fails to address safety concerns and brushes off ethical issues as "beyond the scope of this paper". It mentions potential privacy concerns, but does not detail how these will be addressed. The suggested features, such as watching users sleep, are especially invasive and should be addressed. Physical safety concerns have also not been addressed, such as collision detection or the reliability of the system. This is especially important for technology in the healthcare sector, and should be discussed in a publishable paper.

4 Minor Concerns

In addition to the major concerns about this paper stated above, there are several minor concerns that require revisions. First of all, the paper has many spelling and grammatical mistakes. For this paper to reach a publishable state, all these mistakes, as well as any others within the text, must be resolved. There are too many grammatical inaccuracies to cover in this review, so a non-comprehensive list of grammatical inaccuracies is given as follows:

1. Spelling Mistakes
    - *"(partivuarly repetitive and simple ones)"* → particularly
    - *hospital personell* → personnel
2. Missing words and articles
    - *"Arduino mega 2560 is used for the large number of digital pins (12 alone are required for moving the legs)."* → An Arduino Mega 2560 ...
    - *"Inconvenient tasks like picking up fallen objects or items an elderly has dropped to avoid bending down."* → an elderly person
3. Incorrect plurals
    - *"Disabled and elderly assistant devices follow a similar criteria."* → a similar criterion.
4. Missing punctuation
    - *"No obstacle detection is performed, hence the robot will bump into any objects The robot will stop if it cannot find the marker."* → ... objects. The robot ...
5. Corrupted text
    - *"used.*
      –¿ arco principes of opereation"

In addition to the grammatical errors, the paper is generally poorly worded. It contains many rambling or incomplete sentences and awkward phrasings. For example, *"Due to the nature of robots, however, repetitive tasks without the need for complex decision making are desirable as opposed to ones that require a significant level of decision making (medical diagnosis for instance)."* This sentence is unnecessarily complicated and could be easily rephrased to be more concise. The author also uses redundant definitions such as: *"The robot must be seamless and operate effectively within it's environment even to the extent of operating effectively within an unfamiliar environment that it was not designed for in a way that does not compromise human safety"*. This sentence could be significantly shortened, without the repetition or tautilogical reference to an unfamiliar environment. Thorough proofreading would resolve this concern. The reference usage throughout the paper is also often incorrect or inconsistent. Some examples of this include:

- Empty citation placeholders: *"[] illustrates a robot capable..."*
- Incorrect citation positioning: *"12.5% of the population in the US is elderly and is expected to increase by twice within the coming 5 years.[2]"*
- Inconsistent citation formatting: *"With the rise in technology [1]"*
- Addressing citations directly: *"However, [2] raises some interesting ethical issues regarding robots in daily lives and health care."*
- Unreferenced claims regarding healthcare staffing and demographics

The paper also has multiple inconsistencies in relation to formatting. The paragraph length and spacing are inconsistent throughout the paper. The subsections are formatted strangely; for example, the Approach section starts with a hardware list before entering a very short Overall System subsection. In addition to this, Figure 1 is missing entirely despite the presence of its label. These listed inconsistencies are relatively minor concerns and could be fixed easily by the author.

Overall, while the paper addresses a socially relevant problem area, as it currently stands, it lacks technical rigor, experimental validation, or the clarity of discussion expected of a computer vision research paper. Tightening the scope of the project and defining and executing quantitative experiments would substantially improve the quality of the work.